

CAP BEAR, France

WMO: 077490

Lat: 42.5162N

Lon: 3.1333E

Elev: 87

StdP: 100.28

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.7	3.3	-7.3	2.1	5.0	-5.2	2.4	5.5	30.9	7.7	28.6	8.2	15.1	320	0.750

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.5	29.3	20.6	28.2	20.5	27.1	20.2	23.5	26.0	22.7	25.4	22.1	24.9	6.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.6	24.9	21.9	16.7	24.3	21.1	15.9	23.9	70.3	26.0	67.5	25.4	65.0	24.8	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.8	24.1	21.6	0.2	31.9	1.8	1.5	-1.1	32.9	-2.1	33.8	-3.1	34.6	-4.4	35.7		
			-1.9	24.6	1.7	1.0	-3.1	25.4	-4.2	26.0	-5.1	26.6	-6.4	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	9.4	9.6	11.8	14.0	17.3	21.4	23.8	23.7	20.5	17.4	13.1	10.6
	DBStd	5.76	2.87	3.01	2.65	2.52	2.59	2.64	2.20	2.05	2.28	2.62	2.97	2.84
	HDD10.0	132	45	37	12	2	0	0	0	0	0	1	9	27
	HDD18.3	1354	277	243	202	131	49	4	0	0	6	47	157	239
	CDD10.0	2364	27	27	69	123	227	343	428	426	314	231	103	46
	CDD18.3	544	0	0	0	2	18	97	169	168	71	19	1	0
	CDH23.3	2344	0	0	0	2	35	353	924	863	147	20	0	0
	CDH26.7	333	0	0	0	0	3	48	144	127	10	1	0	0
Wind	WSAvg	8.9	9.9	10.3	9.2	9.2	8.7	8.0	8.3	7.8	8.1	8.3	9.8	9.2
Precipitation	PrecAvg	636	62	46	51	55	54	30	16	28	59	93	79	62
	PrecMax	957	199	131	167	213	128	141	43	70	172	266	240	218
	PrecMin	451	0	1	0	6	8	2	3	2	19	5	13	3
	PrecStd	152	53	36	45	47	38	30	11	17	37	65	70	64
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.5	17.7	20.5	22.4	26.5	29.8	30.8	31.7	28.1	25.7	20.7	18.3
		MCWB	11.8	11.0	12.6	14.6	18.2	20.2	21.1	21.7	19.6	17.8	15.5	12.8
	2%	DB	15.8	15.7	18.2	20.3	24.0	27.9	29.3	29.4	26.1	23.3	19.1	16.5
		MCWB	11.1	10.8	12.1	14.0	17.2	20.0	20.7	21.0	19.3	17.9	15.0	12.1
	5%	DB	14.6	14.5	16.6	19.0	22.2	26.5	28.2	27.9	25.0	22.0	18.0	15.4
		MCWB	10.5	10.2	11.4	13.6	16.2	19.5	20.5	20.9	19.0	17.5	14.7	11.7
	10%	DB	13.4	13.5	15.3	17.7	20.9	25.2	27.2	26.9	23.9	20.7	17.1	14.4
		MCWB	9.8	9.8	11.1	13.0	15.7	19.1	20.3	20.5	18.6	17.1	14.1	11.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	12.7	14.2	16.3	19.3	23.2	24.3	24.7	22.3	20.4	17.5	14.2
		MCDB	15.9	14.9	17.7	19.8	23.6	25.3	27.0	27.0	24.4	22.4	18.4	16.1
	2%	WB	12.1	11.9	13.2	15.2	18.3	21.8	23.3	23.7	21.4	19.5	16.5	13.2
		MCDB	14.4	14.1	15.9	18.5	21.8	24.8	26.0	26.2	23.7	21.3	18.1	15.5
	5%	WB	11.2	11.2	12.5	14.5	17.5	21.0	22.6	23.0	20.7	18.8	15.5	12.4
		MCDB	13.7	13.5	14.9	17.5	20.7	24.6	25.6	25.5	23.1	20.4	17.4	14.8
	10%	WB	10.4	10.5	12.0	13.8	16.8	20.2	21.8	22.3	20.0	18.1	14.5	11.6
		MCDB	12.9	12.9	14.1	16.5	19.7	24.0	25.2	25.0	22.6	19.9	16.5	13.9
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.8	4.9	5.1	5.2	5.5	5.5	5.4	5.2	4.6	4.5	4.4
		MCDBR	5.6	6.1	6.4	6.4	6.6	6.7	6.5	6.2	6.0	5.7	5.2	5.4
		MCWBR	4.0	4.1	4.0	3.8	3.8	3.5	3.9	4.1	4.1	3.8	3.7	3.7
	5% WB	MCDBR	5.0	5.3	5.3	5.6	5.6	5.6	5.6	5.4	5.2	4.6	4.3	4.6
		MCWBR	3.9	4.0	3.8	3.9	3.7	3.6	3.9	4.2	4.1	3.4	3.6	3.7
Clear-Sky Solar Irradiance	taub	0.315	0.329	0.371	0.399	0.411	0.419	0.420	0.416	0.396	0.376	0.339	0.313	
	taud	2.521	2.482	2.382	2.313	2.330	2.342	2.352	2.365	2.404	2.458	2.505	2.527	
	Ebn at Noon	832	875	871	869	865	856	851	843	835	807	791	799	
	Edh at Noon	71	86	107	123	125	124	121	116	104	88	72	65	
All-Sky Solar Radiation	RadAvg	1.83	2.75	3.96	5.00	5.86	6.69	6.67	5.86	4.53	3.08	2.05	1.63	
	RadStd	0.19	0.27	0.32	0.40	0.40	0.32	0.32	0.29	0.26	0.26	0.21	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+46

Nomenclature: See separate page